Present Day, Present Time: AI Exposure and Labor Market Frictions

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Abstract

1 Introduction

2 Background

2.1 Technology, Tasks, and Labor Markets

- **(acemoglu2019automation)** Develops a theoretical model explaining how automation both displaces existing tasks and creates new ones. Highlights the ambiguous effects of technology on labor demand depending on whether new tasks are complements or substitutes for human labor.

- **(ottaviano2013immigration)** Extends the **grossman2008trading<empty citation>** task-trade framework to allow heterogeneous worker types and multiple cutoffs, where different skill groups face different automation (offshoring) thresholds. Originally developed to study offshoring and immigration — not AI — but the formal structure maps directly onto AI-shock analysis: the cutoff mechanism and comparative statics this paper uses are drawn from this framework. Illustrates how a model built in 2013 for a different labor-market shock generalizes to the generative AI context with minimal modification.

- **(autor2025expertise)** Argues that real-world workers are not single-task specialists but hold *bundles* of complementary tasks whose value derives from expertise accumulated over time. Formal model shows that task automation within a bundle can both displace and restructure work depending on whether the automated task is central to the bundle's value. Provides a more nuanced treatment of worker-task assignment than the full-specialization benchmark used in this paper; to the extent workers hold expertise bundles, the wage compression result here is a lower bound and displacement estimates may understate adjustment costs.

2.2 Measuring Exposure to Generative AI

- **(felten2018method)** Introduces a quantitative approach linking progress in distinct AI subfields (from the EFF AI Progress metrics) to human abilities defined in O*NET. Maps AI capabilities $\rightarrow$ O*NET abilities $\rightarrow$ occupations.

- **(felten2021aioe)** Extends the 2018 framework to a forward-looking exposure index using Mechanical Turk surveys to map ten AI applications to 52 O*NET abilities. Aggregates ability-level

exposure (weighted by importance and prevalence) to occupation, industry, and geographic levels. Produces standardized exposure scores such as *AI Occupational Exposure (AIOE)*, *AI Industry Exposure (AIIE)*, and *AI Geography Exposure (AIGE)*, measuring potential technical exposure rather than realized adoption.

- **(webb2020impact)** Constructs a task-technology overlap model matching verbs and nouns from patent text to O*NET task descriptions. Identifies high-skill cognitive tasks (judgment, analysis) as most AI-exposed, contrasting with robotics' focus on routine manual work. Provides a scalable, data-driven mapping of AI technologies to tasks using textual similarity.

- **(eloundou2024gpts)** Uses GPT-4 and human annotators to classify O*NET tasks into categories based on whether LLMs could reduce task completion time by at least 50%. Aggregates task labels to occupation-level exposure scores and produces composite indices. Provides a large-scale, LLM-specific exposure dataset reflecting technical feasibility rather than observed adoption.

- **(handa2025economic)** Analyzes over four million anonymized Claude.ai conversations, mapping them to O*NET tasks and occupations to produce the first large-scale empirical portrait of how AI is *actually used* across the economy — approximating real-world adoption rather than technical feasibility. Finds that AI use is concentrated in software development and writing; roughly 36% of occupations use AI for at least a quarter of their tasks; 57% of interactions suggest augmentation while 43% suggest automation. Represents arguably the most current measure of AI exposure, incorporating observed behavior rather than expert or model annotation. This paper's analysis predates its release and relies on **eloundou2024gpts<empty citation>**; incorporating usage-based measures is a natural direction for future work.

2.3 What the Evidence Shows So Far

- **(hosseini2025seniority)** Examines whether generative AI constitutes a seniority-biased technological change, disproportionately affecting junior relative to senior workers. Combines LinkedIn résumé data (Revelio Labs) with nearly 200 million job postings. Measures firm-level AI adoption by identifying postings for "GenAI integrator" roles responsible for implementing or integrating

AI tools. Employs a difference-in-differences framework comparing junior versus senior employment in adopting versus non-adopting firms before and after 2023. Finds that after ChatGPT's release, junior employment in adopting firms declined by roughly 9–10%, while senior employment remained stable, with effects concentrated in high-exposure occupations. Provides early empirical evidence that generative AI adoption shifts firm workforce composition toward senior labor.

- **(brynjolfsson2025canaries)** Uses high-frequency administrative payroll data from ADP, covering millions of U.S. workers, to document six stylized facts about the early labor market effects of generative AI. Identifies that employment among early-career workers (ages 22–25) in highly AI-exposed occupations has declined by roughly 13% since late 2022, even after controlling for firm-level shocks. Employment for older or less AI-exposed workers remained stable or continued to grow. Shows that employment declines are concentrated in occupations where AI automates rather than augments work, based on task-level measures from the Anthropic Economic Index (**handa2025economic**) and GPT-4 exposure (**eloundou2024gpts**). Finds limited wage effects, suggesting adjustments occurred primarily through employment rather than compensation.

3 Toy Model

3.1 Setup

Production requires a continuum of cognitive tasks indexed $i \in [0, 1]$, ordered by susceptibility to AI automation. Tasks at the low end are routine cognitive tasks easily replicated by large language models; tasks at the high end require human expertise that AI currently struggles to replicate.

Final output Y aggregates tasks via a CES technology:

$$Y = \left[\int_0^1 y(i)^{\frac{\sigma-1}{\sigma}} \, di \right]^{\frac{\sigma}{\sigma-1}}, \qquad \sigma > 0, \tag{1}$$

where $y(i)$ is the quantity of task i used in production.

AI can additionally supply a subset of these tasks. Write $s(i) \geq 0$ for the quantity of task i

that AI contributes, with

$$s'(i) < 0 \text{ for } i < I^*, \qquad s(i) = 0 \text{ for } i \geq I^*, \tag{2}$$

where $I^* \in (0,1)$ is the *AI capability frontier*. Tasks below I^* can be partially supplied by AI, with the contribution strictly decreasing in task complexity; tasks at or above I^* remain beyond AI's reach. As AI improves, I^* shifts rightward and s may increase throughout its support.

Assumption 1 (Small Open Economy)**.** *The output price P_Y is exogenous and fixed.*

Assumption 2 (Full Specialization)**.** *Workers are indexed by ability type $i^* \in [0,1]$. A worker of type i^* fully specializes in task i^*—the task at which they hold comparative advantage over other types. Each task i is therefore supplied exclusively by workers of type $i^* = i$.*

Assumption 3 (Inelastic Labor Supply)**.** *The density of worker types $f(i)$ over $[0,1]$ is fixed and exogenous. Workers do not adjust their type in response to wages, so the pre-AI supply of task i is just $f(i)$ regardless of the task price.*

Competitive firms take P_Y and task prices $\{p_0(i)\}$ as given and minimize the cost of producing Y subject to (1). The equilibrium task prices and wages are derived in Appendix A.

3.2 Pre-AI Wages

Before AI, cost minimization subject to (1) yields equilibrium task prices (derived in Appendix A):

$$p_0(i) = P_Y \cdot \left(\frac{Y_0}{f(i)}\right)^{1/\sigma}, \tag{3}$$

where $Y_0 \equiv \left[\int_0^1 f(i)^{(\sigma-1)/\sigma} di\right]^{\sigma/(\sigma-1)}$ is pre-AI aggregate output, fully determined by the distribution of worker types f via the CES aggregator. In this equilibrium, a worker of type i^* earns:

$$w_0(i^*) = p_0(i^*). \tag{4}$$

Figure 1 illustrates this wage schedule. The figure plots $p_0(i)$ as linearly increasing in i — this is only for visual clarity, the model imposes no explicit restriction on the shape of p_0.

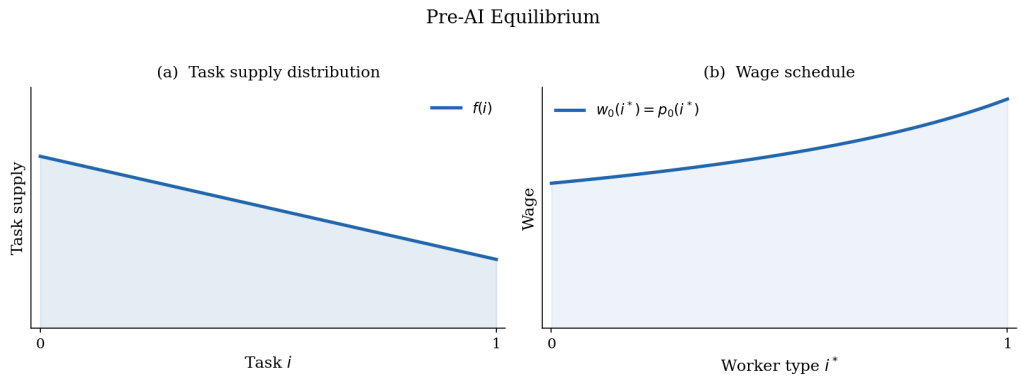

Figure 1: Pre-AI equilibrium wage schedule. Each worker of type i^* earns the competitive task price $w_0(i^*) = p_0(i^*)$. The upward slope is assumed for illustration only; the model places no explicit restriction on the shape of $p_0(i)$.

3.3 Effects of the AI Supply Shock

When AI enters the economy, it augments the supply side of each task market it can serve. The effective supply of task i becomes

$$\bar{y}(i) \;=\; f(i) + s(i), \tag{5}$$

where $s(i) > 0$ for $i < I^*$ (AI contributes to supply) and $s(i) = 0$ for $i \geq I^*$ (beyond AI's capability). Because some task supplies have risen while others have not, the CES aggregator yields a new level of aggregate output

$$\bar{Y} \;=\; \left[\int_0^1 \bar{y}(i)^{\frac{\sigma-1}{\sigma}}\, di\right]^{\frac{\sigma}{\sigma-1}} \;>\; Y_0, \tag{6}$$

and by the same cost-minimization logic as before (Appendix A), the post-AI task price is

$$p_1(i) \;=\; P_Y \left(\frac{\bar{Y}}{\bar{y}(i)}\right)^{1/\sigma}. \tag{7}$$

One channel is set aside: *augmentation* — AI may eventually raise the effective productivity of workers it complements rather than supplements, altering the production technology itself. This is left for future work.

Post-AI wages. Since $w_1(i^*) = p_1(i^*)$, the post-AI wage is read directly from (7):

$$w_1(i^*) \;=\; P_Y \left(\frac{\bar{Y}}{f(i^*) + s(i^*)}\right)^{1/\sigma}. \tag{8}$$

The wage change relative to the pre-AI benchmark $w_0(i^*) = P_Y\big(Y_0/f(i^*)\big)^{1/\sigma}$ is governed by two competing movements: the numerator has grown ($\bar{Y} > Y_0$) because AI adds productive capacity, while the denominator has grown ($f + s > f$) for tasks AI can supply. The interaction of these two forces partitions workers into three groups, examined in turn below.

Group 1: Workers above the frontier ($i^* \geq I^*$). For these workers $s(i^*) = 0$, so the denominator is unchanged. The wage ratio simplifies to

$$\frac{w_1(i^*)}{w_0(i^*)} \;=\; \left(\frac{\bar{Y}}{Y_0}\right)^{1/\sigma} \;>\; 1. \tag{9}$$

These workers *unambiguously benefit*: the economy's expanded output raises the marginal value of every task whose supply has not changed. The gain is uniform across all $i^* \geq I^*$ and increasing in the aggregate size of the AI supply shock.

Group 2: Heavily exposed workers ($i^* < I^*$, large s/f). For workers deep inside $[0, I^*)$, AI supply $s(i^*)$ is large relative to human supply $f(i^*)$. The wage ratio becomes

$$\frac{w_1(i^*)}{w_0(i^*)} \;=\; \left(\frac{\bar{Y}}{Y_0}\right)^{1/\sigma} \cdot \left(\frac{f(i^*)}{f(i^*) + s(i^*)}\right)^{1/\sigma}. \tag{10}$$

The first factor exceeds one (output expansion), but the second factor is strictly less than one and shrinks as s/f grows. A worker experiences wage *compression* whenever

$$\frac{s(i^*)}{f(i^*)} \;>\; \frac{\bar{Y}}{Y_0} - 1, \tag{11}$$

i.e. whenever the proportional supply increase in their task exceeds the proportional increase in aggregate output. Because $\bar{Y}/Y_0$ reflects a CES average over *all* tasks — including the majority where $s = 0$ — while $s(i^*)/f(i^*)$ is concentrated among AI-supplied tasks, condition (11) is satisfied for workers whose tasks are heavily supplemented by AI. These workers face unambiguous wage losses.

Group 3: Mildly exposed workers ($i^* < I^*$, small s/f). Near the frontier, $s(i^*) \to 0$ as $i^* \to I^*$ (by continuity of s). For these workers the second factor in (10) is close to one, and the

output expansion factor dominates:

$$\frac{w_1(i^*)}{w_0(i^*)} \approx \left(\frac{\bar{Y}}{Y_0}\right)^{1/\sigma} > 1 \qquad \text{when } s(i^*)/f(i^*) \text{ is small.}$$

These workers *gain* despite being below I^*. There exists a threshold task $\hat{\imath} < I^*$ at which the two effects exactly offset:

$$\frac{s(\hat{\imath})}{f(\hat{\imath})} = \frac{\bar{Y}}{Y_0} - 1.$$

Workers with $i^* > \hat{\imath}$ (including all of $[I^*, 1]$) experience wage gains; workers with $i^* < \hat{\imath}$ experience wage compression. The boundary between winners and losers therefore lies *strictly below* I^*, not at it.

Displacement. Workers anywhere in the distribution whose post-AI wage falls below a reservation threshold exit employment; displacement is the extensive-margin counterpart of wage compression. As AI capability improves (I^* rises and s increases), the set of compressed workers expands and the threshold $\hat{\imath}$ shifts rightward.

Figure 2 illustrates these effects.

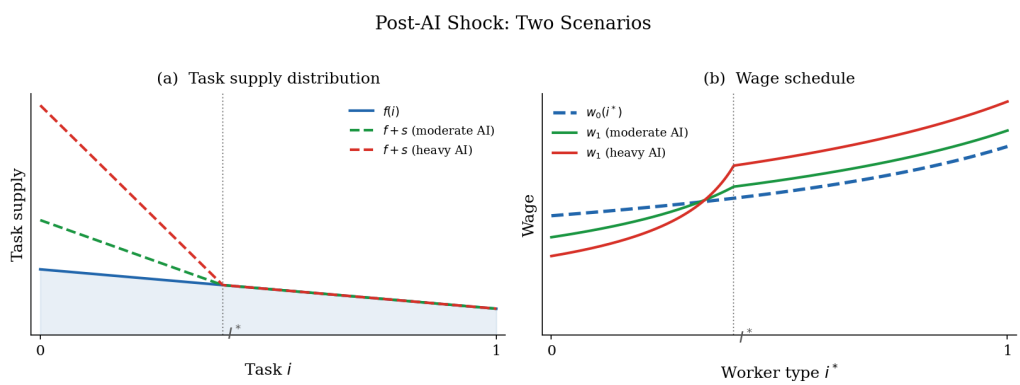

Figure 2: Post-AI shock under two AI supply intensities. Left panel: task supply $f(i)$ (blue) and $f(i) + s(i)$ (dashed). Right panel: pre-AI wage w_0 (dashed blue) vs. post-AI wage w_1 (solid). Workers deep inside $[0, I^*)$ with large s/f see wage compression; workers near I^* with small s/f can still gain from output expansion; workers above I^* gain uniformly.

4 Data

4.1 CPS-ASEC

The primary data source is the Annual Social and Economic Supplement to the Current Population Survey (CPS-ASEC), accessed through IPUMS (**ipumscps2025**). The CPS-ASEC is a nationally representative annual survey that captures labor-force status, occupation, wage income, and detailed demographic characteristics for approximately 100,000 households. I use eight survey waves covering income reference years 2018–2025, spanning both the pre- and post-ChatGPT periods.

I restrict the sample to respondents aged 22–65 who are active in the labor force - either employed or actively seeking work. Workers with less than a high school diploma are excluded. After merging with the AI exposure variable and dropping observations with missing values, the main estimation sample contains 500,669 person-year observations.

Unemployment. The unemployment indicator equals one if a respondent is classified as unemployed by the CPS - not working but actively searching for work (CPS `empstat` codes 21 and 22) - and zero otherwise. The sample mean is 3.4%, roughly consistent with official U.S. unemployment statistics over this period.

Wages. For the wage analysis, I use annual real wage and salary income (`incwage`), deflated to 1982 dollars using the annual average CPI-U from FRED (**fred˙cpiaucsl**). The outcome variable is the natural log of real wages. Observations with zero or missing wage income are excluded from the wage sample (workers who are unemployed or out of the labor force for wage purposes), yielding 464,300 observations.

4.2 AI Exposure

The AI exposure variable is drawn from **eloundou2024gpts<empty citation>**, which constructs occupation-level LLM exposure scores from the O*NET 27.2 database. O*NET provides two complementary task-level resources: 19,265 *occupational tasks* (occupation-specific units of work tied to a particular occupation) and 2,087 *Detailed Work Activities* (DWAs) — broader action categories that may be associated with one or more tasks across occupations.

Exposure rubric. Annotators assess each task or DWA on a three-category rubric, asking whether access to an LLM could reduce the time required to complete it by at least 50% at equivalent quality:

- **E0** (no exposure): the LLM yields no meaningful time reduction, or degrades output quality.
- **E1** (direct exposure): a standard LLM interface—ChatGPT or the OpenAI Playground—reduces task time by at least 50% on its own.
- **E2** (LLM-augmented exposure): the LLM alone cannot meet the threshold, but software built *on top of* the LLM could do so.

Where DWA labels are available, they are first aggregated to the task level before proceeding to the occupation level.

Occupation-level aggregation. For each occupation, O*NET designates each task as *core* (weight $w_t = 2$) or *supplemental* (weight $w_t = 1$). Three summary measures vary how E1 and E2 tasks are scored:

$$\text{Occupation Score} = \frac{\sum_{t \in T_o} w_t \cdot s_t}{\sum_{t \in T_o} w_t}, \tag{12}$$

where $s_t = 1$ for E1 tasks, $s_t = 0.5$ for E2 tasks, and $s_t = 0$ for E0 tasks. The *alpha* score (α_o) sets $s_t = 0$ for all E2 tasks, counting only direct LLM exposure. The *zeta* score (ζ_o) sets $s_t = 1$ for E2 tasks, providing an upper bound. β_o is the weighted middle ground: E2 tasks are discounted by half, reflecting the additional efforts required to build and deploy LLM-powered software on top of the base model.

Annotators. eloundou2024gpts<empty citation> collect two independent sets of annotations. *Human ratings* — produced by the authors and experienced alignment annotators — are the primary benchmark; they were applied to DWAs and aggregated to the task level. *GPT-4 ratings* were obtained by prompting an early GPT-4 version directly on all task/occupation pairs, serving as a validation dataset. The two sources are highly correlated at the occupation level.

I use `dv_rating_beta`: the *GPT-4-annotated* β score. Applying a single model uniformly across all task/occupation pairs ensures internal consistency. The cost is potential overconfidence in

GPT-4's assessment of its own capabilities. β is preferred to α (which ignores the ecosystem of LLM-powered tools already deployed or in development at the time of study) and ζ (which assumes a hypothetically well-developed AI infrastructure). β is therefore the most economically meaningful measure for the post-ChatGPT period.

The variable ranges from approximately 0 (low exposure, e.g., Construction and Extraction) to 0.78 (high exposure, e.g., Computer and Mathematical, Legal, Business and Financial Operations), with a sample standard deviation of roughly 0.17.Because $\beta_o \in [0, 1]$, all regression coefficients are interpreted per 0.1-unit increase throughout. The variable is time-invariant and assigned at the occupational level.

Because CPS-ASEC codes occupations using a Census-based scheme which is distinct from the SOC occupation codes used by **eloundou2024gpts<empty citation>**, I construct a hierarchical crosswalk: from Census 2010 occupation codes to 6-digit 2010 SOC codes, then from 2010 SOC to 2018 SOC using the BLS official crosswalk (**bls˙soc˙crosswalk˙2018**). Where exact 6-digit matches are unavailable, exposure scores are aggregated at the 5-digit and 4-digit SOC levels, taking group means. This procedure retains nearly all observations in the sample.

Potential experience. For the experience heterogeneity specification, I impute potential labor-market experience as max(age − years of schooling − 6, 0), with years of schooling assigned from the CPS education recode.

4.3 Descriptive Statistics

Figure 3 plots the kernel density of the AI exposure score β_o across workers. Most workers cluster at low-to-moderate exposure, with a long right tail among which are mainly cognitively heavy occupations.

To ground the exposure measure, Table 1 lists the five most exposed occupations and Table 2 the five least exposed, drawn directly from **eloundou2024gpts<empty citation>**. Programmers, legal professionals, and financial analysts anchor the high end; construction, farming, and building maintenance anchor the low end.

Figure 4 overlays the exposure distributions for college-educated and non-college workers. College workers are shifted rightward — they are more exposed on average — but the two distribu-

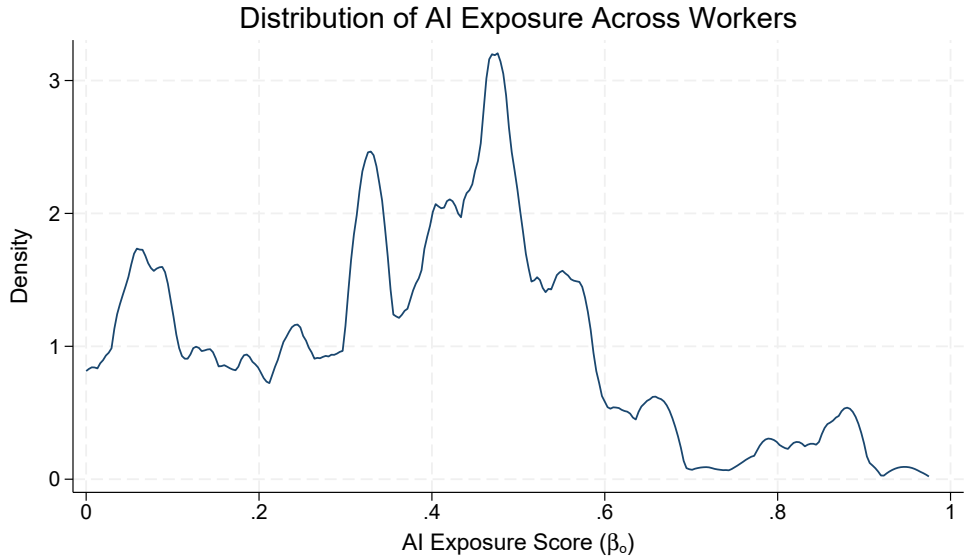

Figure 3: Distribution of AI exposure (β_o) across workers. Each observation is a worker-year; the density reflects the worker-level distribution of occupational exposure scores.

Table 1: Most exposed occupations (top 5 by β_o).

Occupation	Exposure Score
Mathematicians	1.000
Proofreaders and Copy Markers	0.975
Blockchain Engineers	0.971
Correspondence Clerks	0.964
Court Reporters and Simultaneous Captioners	0.958

Table 2: Least exposed occupations (bottom 5 by β_o).

Occupation	Exposure Score
Cement Masons and Concrete Finishers	0.000
Packaging and Filling Machine Operators and Tenders	0.000
Motorcycle Mechanics	0.000
Electrical Power-Line Installers and Repairers	0.000
Helpers–Pipelayers, Plumbers, Pipefitters, and Steamfitters	0.000

tions overlap substantially. This overlap is important for the education heterogeneity analysis in Section 6: any asymmetry in outcomes between the two groups cannot be attributed simply to differential exposure.

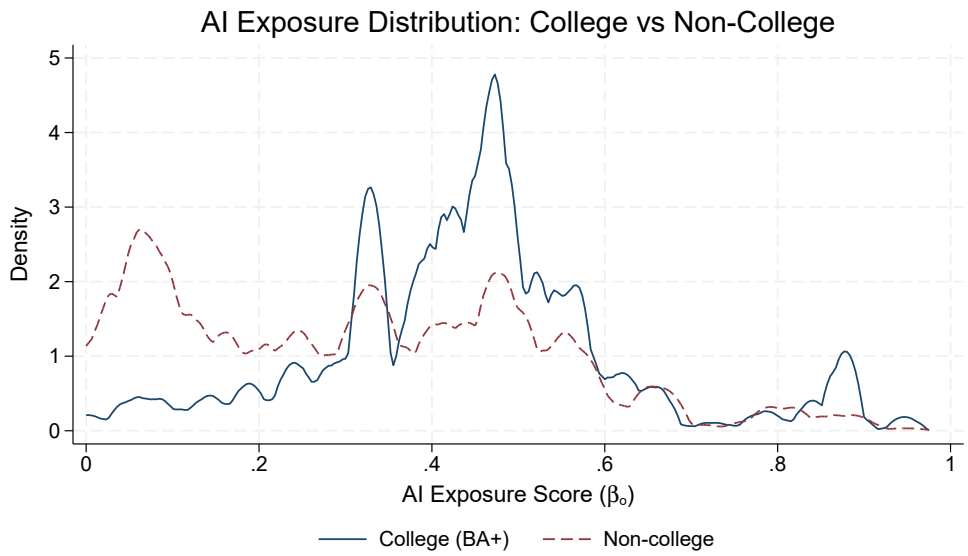

Figure 4: Distribution of AI exposure (β_o): college-educated workers (BA or above) vs. non-college workers (high school or some college).

Finally, Table 3 ranks all 22 major CPS occupation groups by mean AI exposure and reports the share of college-educated workers in each group. The most exposed groups (computer and mathematical, office and administrative support, architecture and engineering) span a wide range of education intensity, reinforcing the point that AI exposure is not simply a proxy for skill level.

5 Empirical Strategy

5.1 Main Specification

The core empirical question is whether outcomes for workers in high-exposure occupations moved differently from outcomes in low-exposure occupations after ChatGPT's release, relative to how they moved before. The estimating equation is:

$$Y_{it} = \alpha + \sum_{t \neq 2022} \beta_t \left(\mathbf{1}[t] \times \text{AIExposure}_i \right) + \text{lower-order terms} + X_{it}' \Gamma + \varepsilon_{it}, \tag{13}$$

Table 3: Occupation groups ranked by AI exposure (β_o), with share of college-educated workers.

Occupation Group	AI Exposure (β_o)	% College
Computer and Mathematical Occupations	0.791	76.3
Office and Administrative Support Occupations	0.576	28.9
Architecture and Engineering Occupations	0.511	74.9
Business and Financial Operations Occupations	0.510	71.5
Legal Occupations	0.486	82.7
Arts, Design, Entertainment, Sports, and Media Occupations	0.484	66.7
Management Occupations	0.461	60.4
Life, Physical, and Social Science Occupations	0.454	88.0
Sales and Related Occupations	0.442	36.7
Educational Instruction and Library Occupations	0.350	80.8
Healthcare Practitioners and Technical Occupations	0.348	66.6
Community and Social Service Occupations	0.301	78.2
Protective Service Occupations	0.288	33.8
Transportation and Material Moving Occupations	0.212	14.3
Healthcare Support Occupations	0.182	18.5
Personal Care and Service Occupations	0.181	22.7
Production Occupations	0.173	13.5
Food Preparation and Serving Related Occupations	0.162	15.3
Installation, Maintenance, and Repair Occupations	0.125	11.5
Construction and Extraction Occupations	0.117	10.8
Farming, Fishing, and Forestry Occupations	0.094	17.0
Building and Grounds Cleaning and Maintenance Occupations	0.052	12.0

where Y_{it} is either the unemployment indicator or log wages for worker i in year t; AIExposure_i is the worker i's occupation's AI exposure score; and year 2022 is the omitted category. The lower-order terms include year fixed effects and the main effect of AI exposure. The β_t coefficients trace, year by year, whether high-exposure workers diverge from low-exposure workers relative to the 2022 baseline. If AI caused labor-market disruption after its diffusion, we expect the post-2022 β_t to be positive (for unemployment) or negative (for wages), while the pre-2022 β_t should hover near zero.

Four progressively stricter specifications are estimated: (1) no additional controls, (2) sex fixed effects, (3) sex and education fixed effects, and (4) sex, education, and state×year two-way fixed effects. Specification (4) absorbs state-specific time shocks and is the preferred specification. Standard errors are clustered at the occupation level across all specifications.

5.2 Heterogeneity: Experience and Education

Experience. To examine whether the effect varies by seniority of the worker, I supplement the main specification with a triple interaction:

$$Y_{it} = \alpha + \sum_{t \neq 2022} \delta_t \left(\mathbf{1}[t] \times \text{Experience}_i \times \text{AIExposure}_i \right) + \text{lower-order terms} + X'_{it}\Gamma + \varepsilon_{it}. \tag{14}$$

A negative post-2022 δ_t would indicate that more-experienced workers are relatively shielded from the influence of AI; a positive value would indicate they are disproportionately affected.

Education. For the education dimension I divide the sample into *college-educated* workers (BA or above, educ > 110) and *non-college* workers (high school or some college, educ ≤ 110) and estimate the main specification (13) separately within each group:

$$Y_{it}^{g} = \alpha^{g} + \sum_{t \neq 2022} \beta_t^{g} \left(\mathbf{1}[t] \times \text{AIExposure}_i \right) + \text{lower-order terms} + X'_{it}\Gamma^{g} + \varepsilon_{it}^{g}, \quad g \in \{\text{college, non-college}\}. \tag{15}$$

Each subsample has its own baseline, trend, and exposure gradient, and the two coefficient paths $\{\hat{\beta}_t^{\text{col}}\}$ and $\{\hat{\beta}_t^{\text{noncol}}\}$ can be compared directly.

6 Results and Interpretation

6.1 Unemployment

Table C1 and Figure 5 present the unemployment results. The pattern is clear and remarkably stable across all four specifications (see Figures B1–B3 for specifications (1)–(3)).

In the pre-ChatGPT period, coefficients for 2018 and 2019 are statistically indistinguishable from zero. The pandemic years (2020–2021) present a noticable negative: AI-exposed occupations experienced *lower* unemployment during COVID-19, with coefficients of -0.0213^{***} in 2020 and -0.0353^{***} in 2021. This reflects the simple fact that high-exposure occupations - legal, financial, managerial, computational - were largely remote / hybrid -compatible and thus are partly shielded from the layoffs that consentrated in the in-person service occupations.

The post-2022 reversal is the central finding. The 2023 coefficient is small and insignificant

(+0.002), consistent with a natural lag between GPT model release and meaningful workforce adjustment. By 2024, the coefficient rises to $+0.0205^{***}$ and remains at $+0.0180^{***}$ in 2025. A 0.1 increase in exposure is associated with a 0.2 percentage point increase in unemployment probability, corresponding to roughly a 6% increase relative to the 3.4% baseline. This effect is robust across all four specifications, including the most demanding one that absorbs state-specific aggregate shocks.

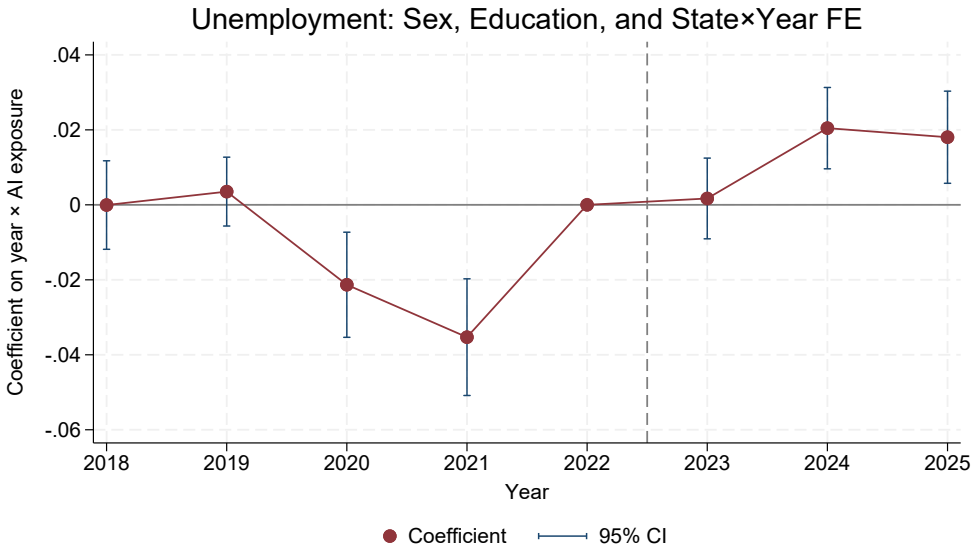

Figure 5: Unemployment: DiD coefficients (year × AI exposure), preferred specification (sex, education, and state×year fixed effects). Year 2022 is the baseline ($\hat{\beta}_{2022} = 0$). Error bars are 95% CI, SEs clustered at the occupation level. Specifications (1)–(3): Figures B1–B3.

6.2 Wages

Table C2 presents the log-wage results. Pre-ChatGPT coefficients are uniformly small and statistically insignificant across all four specifications, confirming parallel wage trends. In the post-2022 period, the naive and sex-FE specifications yield 2025 coefficients of -0.036 and -0.046, neither significant at conventional levels. Only the most demanding specification — absorbing sex, education, and state×year fixed effects — produces a statistically significant 2025 compression of -0.059^{**}. No significant effect appears in 2023 or 2024 under any specification. This weak and late-emerging signal suggests that wage adjustment, if any, lags behind the employment adjustment above and requires absorbing state-level aggregate shocks to emerge clearly.

6.3 Who Bears the Burden: Education Heterogeneity

Unemployment. Splitting the sample by education reveals a symmetric picture for unemployment. Both college-educated workers (BA or above; baseline unemployment rate 2.2%) and non-college workers (high school or some college; baseline 4.3%) in AI-exposed occupations experienced comparable increases in unemployment risk after 2022. For college workers, the post-2022 coefficients are $+0.018^{**}$ in 2024 and $+0.018^{**}$ in 2025, stable across all four specifications. For non-college workers, the corresponding estimates are $+0.020^{***}$ in 2024 and $+0.014^{*}$ in 2025. The point estimates are similar in magnitude and the confidence intervals overlap. Hence there is no statistically significant difference in the unemployment response across education groups. Regarding pre-2022 trends: the 2018 and 2019 coefficients are near zero and insignificant for both groups. The 2020–2021 coefficients are significantly negative in both subsamples - mirroring the main-spec finding that high-exposure occupations shed less unemployment during COVID-19 - and should be read as a replication of that pattern within each education group rather than as evidence against identification. Specifications (1)–(3) yield nearly identical results (Figures B4–B6).

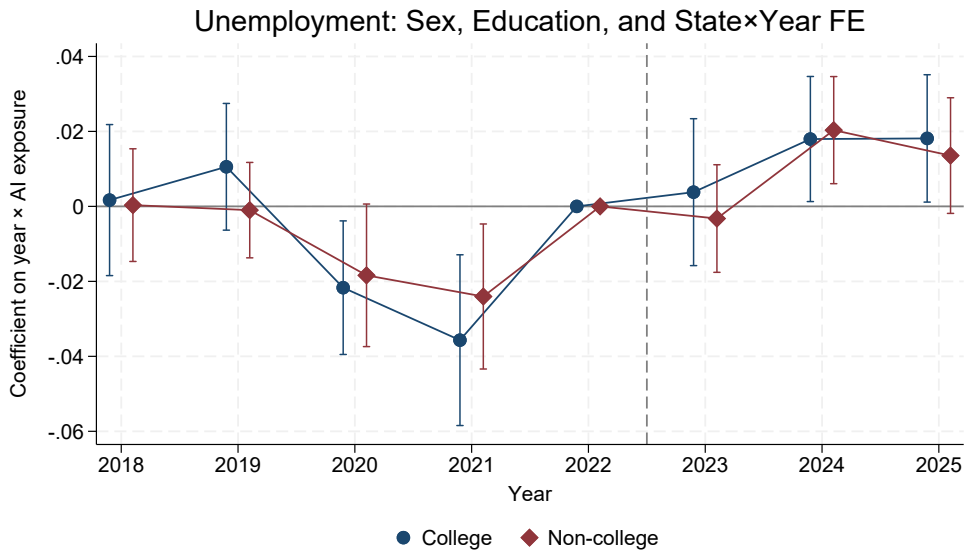

Figure 6: Unemployment: DiD coefficients (year × AI exposure), college (navy circles) vs. non-college workers (maroon diamonds), preferred specification (sex, education, and state×year FE). Year 2022 is the baseline ($\hat{\beta}_{2022} \equiv 0$). Error bars are 95% CI, clustered at the occupation level. Horizontal jitter ±0.1 years for readability. Specifications (1)–(3): Figures B4–B6.

Log Wages. The wage results reveal a divergence across education groups. For college-educated workers, all post-2022 wage coefficients are positive and statistically insignificant: 2023 (+0.066), 2024 (+0.060), 2025 (−0.001). There is no evidence of wage compression among college workers in AI-exposed occupations. For non-college workers, a significant divergence emerges in 2025: the coefficient is -0.072^{*} in the naive specification, rising in magnitude and significance to -0.090^{**}, -0.090^{**}, and -0.094^{***} as progressively richer fixed effects are added. The preferred specification — absorbing sex, education, and state×year fixed effects — yields the strongest estimate, -0.094^{***}. Pre-2022 coefficients are uniformly insignificant for both groups. The education split thus reveals a potential asymmetry: the unemployment burden of AI exposure is broadly shared, but the wage compression materializing in 2025 is concentrated among non-college workers and grows more pronounced as state-level aggregate shocks are controlled for. Specifications (1)–(3): Figures B7–B9.

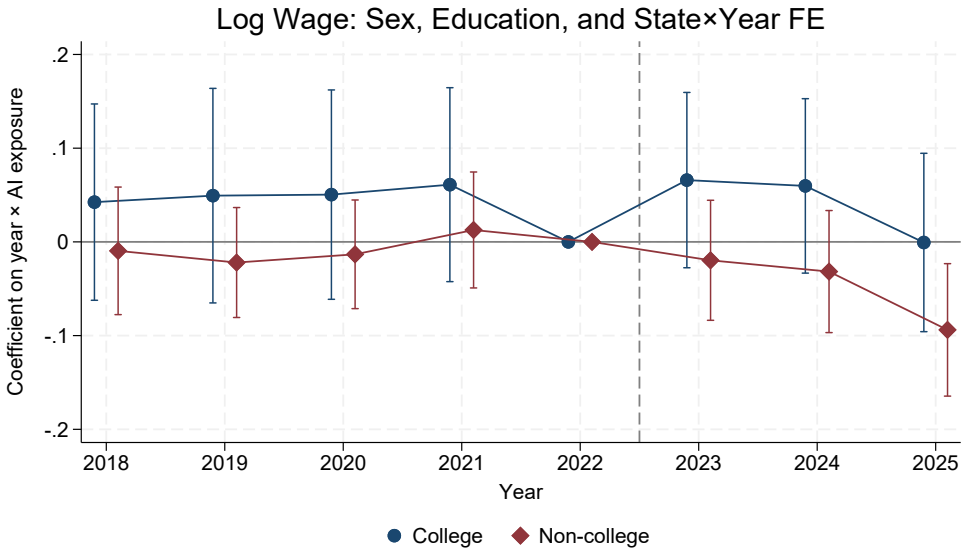

Figure 7: Log wage: DiD coefficients (year × AI exposure), college (navy circles) vs. non-college workers (maroon diamonds), preferred specification (sex, education, and state×year FE). Year 2022 is the baseline. The non-college 2025 compression (-0.094^{***}) stands in contrast to the flat college path. Error bars are 95% CI, clustered at the occupation level. Specifications (1)–(3): Figures B7–B9.

6.4 Experience Heterogeneity

Unemployment. Table C7 reports the unemployment experience-heterogeneity results. The triple-interaction coefficients on year $\times$ experience $\times$ AI exposure are economically small and statistically insignificant across all post-2022 years and all three specifications in both subsamples. For college workers, 2024 and 2025 coefficients are approximately $+0.0007$ with standard errors of similar magnitude. For non-college workers, post-2022 estimates are even smaller (around ± 0.0002) and similarly insignificant. Within either education group, there is no evidence that more-experienced workers in AI-exposed occupations faced systematically different unemployment risk.

Log Wages. The log-wage experience results are largely null: post-2022 triple-interaction coefficients are uniformly small and statistically insignificant across all four specifications for college workers. The borderline-significant 2025 estimate in column (3) (-0.0076^{*}) disappears once state$\times$year fixed effects are added (column (4): -0.0072, n.s.). For non-college workers, the 2023 coefficient of -0.0043^{**} persists across all four specifications, suggesting that more-experienced non-college workers in AI-exposed occupations faced greater wage compression immediately after ChatGPT's release, but the effect does not persist into 2024 or 2025. Interpretation is complicated by significant pre-period coefficients in 2019 (-0.0042^{**}) and 2021 (-0.0065^{***}) for the non-college subsample, indicating that the experience–AI-wage relationship was not flat before 2022 and the 2023 signal cannot be attributed cleanly to the post-ChatGPT treatment.

7 Limitations

8 Conclusion

A Derivation of Wage Schedules

Step 1. Cost minimization $\rightarrow$ FOC: $p_0(i) = \mu \cdot (Y/y(i))^{1/\sigma}, \quad \mu = P_Y$ in competitive equilibrium.

Step 2. Task demand: $y^d(i) = Y(P_Y/p_0(i))^\sigma$.

Step 3. Market clearing $y^d(i) = f(i)$:

$$p_0(i) = P_Y \left(\frac{Y}{f(i)} \right)^{1/\sigma}.$$

Step 4. Plug $y(i) = f(i)$ into CES $\rightarrow$ output is pinned:

$$Y_0 = \left[\int_0^1 f(i)^{(\sigma-1)/\sigma} \, di \right]^{\sigma/(\sigma-1)}.$$

Y_0 depends only on worker distribution f — not on prices. ■

Step 5. Post-AI: effective supply $\bar{y}(i) = f(i) + s(i)$. Repeat Steps 1–4 with $\bar{y}$ in place of f:

$$\bar{Y} = \left[\int_0^1 (f(i) + s(i))^{(\sigma-1)/\sigma} \, di \right]^{\sigma/(\sigma-1)} > Y_0, \qquad p_1(i) = P_Y \left(\frac{\bar{Y}}{f(i) + s(i)} \right)^{1/\sigma}. \quad ■$$

B Regression Tables

Table C1: Unemployment regression results. Dependent variable: binary unemployment indicator. Year 2022 is the omitted baseline. Columns (1)–(4): no fixed effects, sex FE, sex + education FE, sex + education + state×year FE. Standard errors clustered at the occupation level. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Dependent Variable:	Unemployment			
	(1)	(2)	(3)	(4)
2018 × AI exposure	-0.000205	-0.000192	-0.000555	-0.0000526
	(0.00597)	(0.00598)	(0.00596)	(0.00601)
2019 × AI exposure	0.00391	0.00390	0.00351	0.00353
	(0.00474)	(0.00473)	(0.00469)	(0.00467)
2020 × AI exposure	-0.0213***	-0.0213***	-0.0218***	-0.0213***
	(0.00723)	(0.00723)	(0.00722)	(0.00714)
2021 × AI exposure	-0.0335***	-0.0335***	-0.0340***	-0.0353***
	(0.00793)	(0.00793)	(0.00792)	(0.00792)
2023 × AI exposure	0.00202	0.00199	0.00193	0.00170
	(0.00559)	(0.00557)	(0.00556)	(0.00548)
2024 × AI exposure	0.0206***	0.0205***	0.0206***	0.0205***
	(0.00552)	(0.00551)	(0.00551)	(0.00552)
2025 × AI exposure	0.0194***	0.0194***	0.0196***	0.0180***
	(0.00639)	(0.00639)	(0.00635)	(0.00625)
Mean of dependent variable	0.034	0.034	0.034	0.034
Sex FE	No	Yes	Yes	Yes
Education FE	No	No	Yes	Yes
State × Year FE	No	No	No	Yes
SE clustered at	Occupation	Occupation	Occupation	Occupation
Observations	500,669	500,669	500,669	500,669

Table C2: Log wage regression results. Dependent variable: natural log of annual real wages (1982 dollars). Year 2022 is the omitted baseline. Columns (1)–(4): no fixed effects, sex FE, sex + education FE, sex + education + state×year FE. Standard errors clustered at the occupation level. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Dependent Variable:	Log Wage			
	(1)	(2)	(3)	(4)
2018 × AI exposure	-0.0228	-0.0152	-0.00423	0.00158
	(0.0365)	(0.0319)	(0.0301)	(0.0301)
2019 × AI exposure	-0.0236	-0.0277	-0.0151	-0.0134
	(0.0338)	(0.0298)	(0.0273)	(0.0272)
2020 × AI exposure	-0.0145	-0.0230	-0.0115	-0.0107
	(0.0305)	(0.0278)	(0.0258)	(0.0258)
2021 × AI exposure	0.0367	0.0388	0.0518*	0.0488*
	(0.0276)	(0.0263)	(0.0265)	(0.0265)
2023 × AI exposure	0.0227	0.00733	0.00884	0.00835
	(0.0316)	(0.0308)	(0.0284)	(0.0286)
2024 × AI exposure	0.00831	-0.00560	-0.00865	-0.00727
	(0.0314)	(0.0301)	(0.0265)	(0.0266)
2025 × AI exposure	-0.0356	-0.0463	-0.0587**	-0.0592**
	(0.0309)	(0.0282)	(0.0262)	(0.0264)
Mean of dependent variable	9.733	9.733	9.733	9.733
Sex FE	No	Yes	Yes	Yes
Education FE	No	No	Yes	Yes
State × Year FE	No	No	No	Yes
SE clustered at	Occupation	Occupation	Occupation	Occupation
Observations	464,300	464,300	464,300	464,300

Table C3: Unemployment — college subsample. Dependent variable: binary unemployment indicator. Split-sample regression on college-educated workers (BA or above). Year 2022 is the omitted baseline. Columns (1)–(4) as above. Standard errors clustered at the occupation level. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Dependent Variable:	Unemployment			
	(1)	(2)	(3)	(4)
2018 × AI exposure	0.00152	0.00153	0.00171	0.00168
	(0.0101)	(0.0101)	(0.0101)	(0.0103)
2019 × AI exposure	0.0109	0.0109	0.0110	0.0106
	(0.00862)	(0.00862)	(0.00861)	(0.00860)
2020 × AI exposure	-0.0217**	-0.0217**	-0.0215**	-0.0217**
	(0.00906)	(0.00907)	(0.00905)	(0.00907)
2021 × AI exposure	-0.0340***	-0.0340***	-0.0341***	-0.0357***
	(0.0116)	(0.0116)	(0.0116)	(0.0116)
2023 × AI exposure	0.00467	0.00465	0.00474	0.00380
	(0.0103)	(0.0104)	(0.0103)	(0.00997)
2024 × AI exposure	0.0189**	0.0189**	0.0189**	0.0180**
	(0.00860)	(0.00860)	(0.00857)	(0.00849)
2025 × AI exposure	0.0211**	0.0211**	0.0212**	0.0181**
	(0.00867)	(0.00867)	(0.00871)	(0.00865)
Mean of dependent variable	0.022	0.022	0.022	0.022
Sex FE	No	Yes	Yes	Yes
Education FE	No	No	Yes	Yes
State × Year FE	No	No	No	Yes
SE clustered at	Occupation	Occupation	Occupation	Occupation
Observations	221,543	221,543	221,543	221,543

Table C4: Unemployment — non-college subsample. Dependent variable: binary unemployment indicator. Split-sample regression on non-college workers (high school or some college). Year 2022 is the omitted baseline. Columns (1)–(4) as above. Standard errors clustered at the occupation level. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Dependent Variable:	Unemployment			
	(1)	(2)	(3)	(4)
2018 × AI exposure	0.000396	0.000378	0.000442	0.000338
	(0.00764)	(0.00764)	(0.00764)	(0.00765)
2019 × AI exposure	-0.000346	-0.000306	-0.000366	-0.00100
	(0.00662)	(0.00661)	(0.00662)	(0.00648)
2020 × AI exposure	-0.0187*	-0.0187*	-0.0187*	-0.0184*
	(0.00965)	(0.00966)	(0.00968)	(0.00968)
2021 × AI exposure	-0.0233**	-0.0232**	-0.0232**	-0.0240**
	(0.00992)	(0.00992)	(0.00993)	(0.00984)
2023 × AI exposure	-0.00271	-0.00264	-0.00261	-0.00324
	(0.00734)	(0.00732)	(0.00736)	(0.00730)
2024 × AI exposure	0.0207***	0.0208***	0.0209***	0.0203***
	(0.00730)	(0.00729)	(0.00730)	(0.00727)
2025 × AI exposure	0.0135*	0.0136*	0.0136*	0.0136*
	(0.00791)	(0.00790)	(0.00789)	(0.00785)
Mean of dependent variable	0.043	0.043	0.043	0.043
Sex FE	No	Yes	Yes	Yes
Education FE	No	No	Yes	Yes
State × Year FE	No	No	No	Yes
SE clustered at	Occupation	Occupation	Occupation	Occupation
Observations	279,126	279,126	279,126	279,126

Table C5: Log wage — college subsample. Dependent variable: log annual real wages (1982 dollars). Split-sample regression on college-educated workers. Year 2022 is the omitted baseline. Columns (1)–(4) as above. Standard errors clustered at the occupation level. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Dependent Variable:	Log Wage			
	(1)	(2)	(3)	(4)
2018 × AI exposure	0.0512	0.0560	0.0426	0.0424
	(0.0589)	(0.0565)	(0.0534)	(0.0533)
2019 × AI exposure	0.0578	0.0566	0.0538	0.0493
	(0.0624)	(0.0607)	(0.0593)	(0.0583)
2020 × AI exposure	0.0672	0.0653	0.0564	0.0504
	(0.0587)	(0.0554)	(0.0555)	(0.0568)
2021 × AI exposure	0.0505	0.0667	0.0672	0.0611
	(0.0535)	(0.0514)	(0.0510)	(0.0527)
2023 × AI exposure	0.0847	0.0783	0.0737	0.0659
	(0.0528)	(0.0495)	(0.0472)	(0.0476)
2024 × AI exposure	0.0690	0.0637	0.0619	0.0598
	(0.0544)	(0.0519)	(0.0474)	(0.0474)
2025 × AI exposure	0.000340	0.00967	0.00551	-0.000640
	(0.0500)	(0.0489)	(0.0472)	(0.0484)
Mean of dependent variable	10.054	10.054	10.054	10.054
Sex FE	No	Yes	Yes	Yes
Education FE	No	No	Yes	Yes
State × Year FE	No	No	No	Yes
SE clustered at	Occupation	Occupation	Occupation	Occupation
Observations	208,928	208,928	208,928	208,928

Table C6: Log wage — non-college subsample. Dependent variable: log annual real wages (1982 dollars). Split-sample regression on non-college workers. Year 2022 is the omitted baseline. Columns (1)–(4) as above. Standard errors clustered at the occupation level. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Dependent Variable:	Log Wage			
	(1)	(2)	(3)	(4)
2018 × AI exposure	-0.0127	-0.00914	-0.0105	-0.00956
	(0.0394)	(0.0353)	(0.0347)	(0.0346)
2019 × AI exposure	-0.0108	-0.0226	-0.0228	-0.0220
	(0.0350)	(0.0304)	(0.0301)	(0.0299)
2020 × AI exposure	-0.00245	-0.0129	-0.0140	-0.0133
	(0.0327)	(0.0299)	(0.0292)	(0.0295)
2021 × AI exposure	0.0174	0.0158	0.0145	0.0127
	(0.0332)	(0.0325)	(0.0321)	(0.0315)
2023 × AI exposure	-0.00238	-0.0178	-0.0181	-0.0196
	(0.0334)	(0.0331)	(0.0325)	(0.0326)
2024 × AI exposure	-0.0124	-0.0308	-0.0319	-0.0317
	(0.0362)	(0.0342)	(0.0334)	(0.0331)
2025 × AI exposure	-0.0721*	-0.0889**	-0.0904**	-0.0939***
	(0.0373)	(0.0371)	(0.0364)	(0.0360)
Mean of dependent variable	9.471	9.471	9.471	9.471
Sex FE	No	Yes	Yes	Yes
Education FE	No	No	Yes	Yes
State × Year FE	No	No	No	Yes
SE clustered at	Occupation	Occupation	Occupation	Occupation
Observations	255,372	255,372	255,372	255,372

Table C7: Unemployment experience heterogeneity. Dependent variable: binary unemployment indicator. Reported coefficients are on the triple interaction year × potential experience × AI exposure. Full sample. Columns (1)–(4) as above. Standard errors clustered at the occupation level. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Dependent Variable:	Unemployment			
	(1)	(2)	(3)	(4)
2018 × Experience × AI exposure	0.000414	0.000414	0.000408	0.000414
	(0.000366)	(0.000366)	(0.000368)	(0.000367)
2019 × Experience × AI exposure	0.000914***	0.000914***	0.000895***	0.000903***
	(0.000346)	(0.000346)	(0.000342)	(0.000340)
2020 × Experience × AI exposure	0.000568	0.000567	0.000557	0.000548
	(0.000350)	(0.000350)	(0.000344)	(0.000342)
2021 × Experience × AI exposure	0.000711**	0.000710**	0.000718**	0.000736**
	(0.000356)	(0.000356)	(0.000357)	(0.000353)
2023 × Experience × AI exposure	0.000465	0.000463	0.000446	0.000434
	(0.000372)	(0.000372)	(0.000373)	(0.000373)
2024 × Experience × AI exposure	0.0000393	0.0000381	0.0000590	0.0000773
	(0.000387)	(0.000387)	(0.000386)	(0.000387)
2025 × Experience × AI exposure	0.000365	0.000364	0.000368	0.000364
	(0.000412)	(0.000412)	(0.000410)	(0.000410)
Mean of dependent variable	0.034	0.034	0.034	0.034
Sex FE	No	Yes	Yes	Yes
Education FE	No	No	Yes	Yes
State × Year FE	No	No	No	Yes
SE clustered at	Occupation	Occupation	Occupation	Occupation
Observations	500,669	500,669	500,669	500,669

Table C8: Log wage experience heterogeneity. Dependent variable: log annual real wages. Reported coefficients are on the triple interaction year × potential experience × AI exposure. Full sample. Columns (1)–(4) as above. Standard errors clustered at the occupation level. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Dependent Variable:	Log Wage			
	(1)	(2)	(3)	(4)
2018 × Experience × AI exposure	0.000144	-0.0000706	-0.000328	-0.000264
	(0.00202)	(0.00185)	(0.00166)	(0.00165)
2019 × Experience × AI exposure	-0.00250	-0.00273	-0.00247*	-0.00245
	(0.00186)	(0.00166)	(0.00149)	(0.00150)
2020 × Experience × AI exposure	-0.000664	-0.00126	-0.00138	-0.00133
	(0.00200)	(0.00183)	(0.00162)	(0.00163)
2021 × Experience × AI exposure	-0.00351**	-0.00385**	-0.00426***	-0.00439***
	(0.00168)	(0.00156)	(0.00143)	(0.00146)
2023 × Experience × AI exposure	-0.00230	-0.00293**	-0.00287**	-0.00292**
	(0.00159)	(0.00142)	(0.00132)	(0.00132)
2024 × Experience × AI exposure	-0.00139	-0.00196	-0.00237	-0.00234
	(0.00188)	(0.00180)	(0.00168)	(0.00170)
2025 × Experience × AI exposure	-0.00192	-0.00245	-0.00266	-0.00261
	(0.00185)	(0.00175)	(0.00161)	(0.00160)
Mean of dependent variable	9.733	9.733	9.733	9.733
Sex FE	No	Yes	Yes	Yes
Education FE	No	No	Yes	Yes
State × Year FE	No	No	No	Yes
SE clustered at	Occupation	Occupation	Occupation	Occupation
Observations	464,300	464,300	464,300	464,300

C Additional Figures

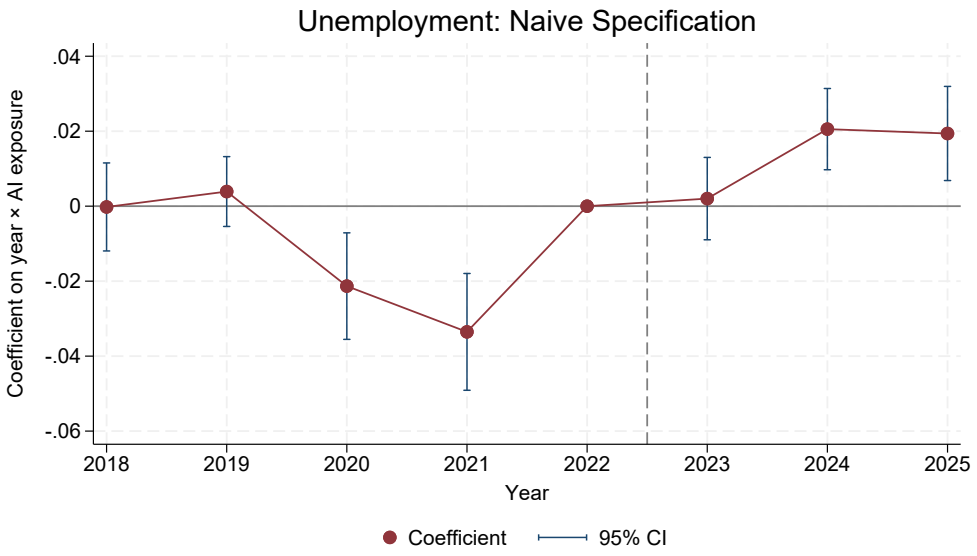

Figure B1: Unemployment: DiD coefficients (year × AI exposure), naive specification (no controls). Year 2022 is the baseline ($\hat{\beta}_{2022} \equiv 0$). Error bars are 95% CI, clustered at the occupation level.

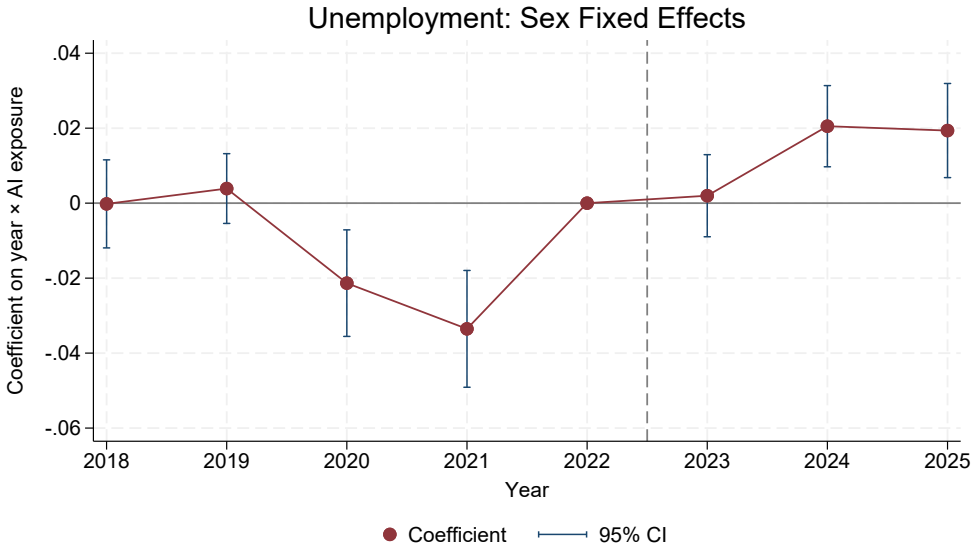

Figure B2: Unemployment: DiD coefficients, sex fixed effects. Year 2022 is the baseline. Error bars are 95% CI, clustered at the occupation level.

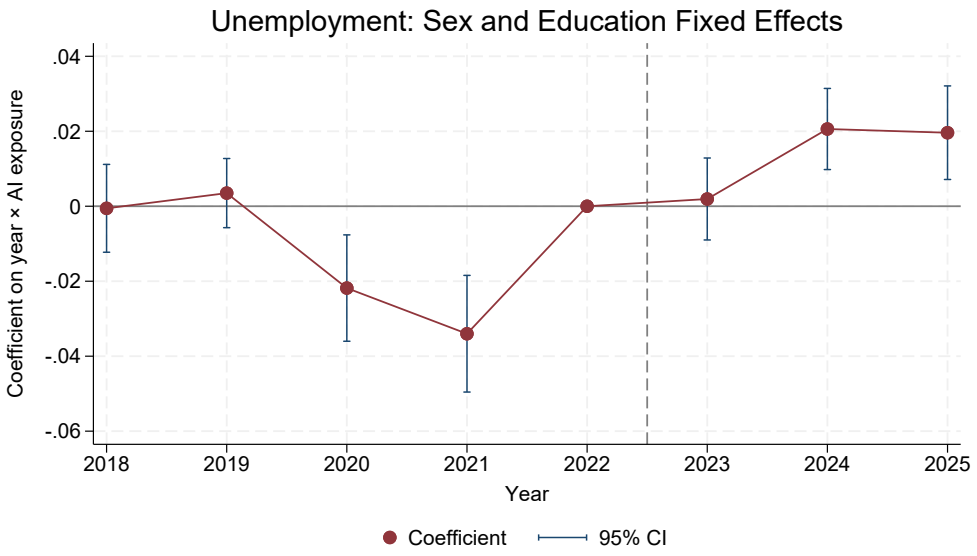

Figure B3: Unemployment: DiD coefficients, sex and education fixed effects. Year 2022 is the baseline. Error bars are 95% CI, clustered at the occupation level.

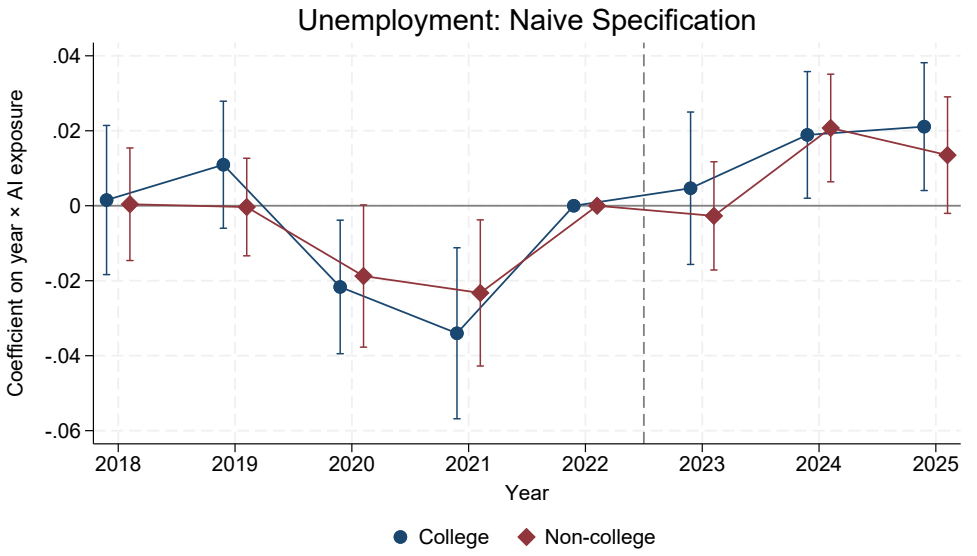

Figure B4: Unemployment: DiD coefficients, college (navy circles) vs. non-college (maroon diamonds), naive specification. Year 2022 is the baseline. Error bars are 95% CI, clustered at the occupation level. Horizontal jitter ±0.1 years for readability.

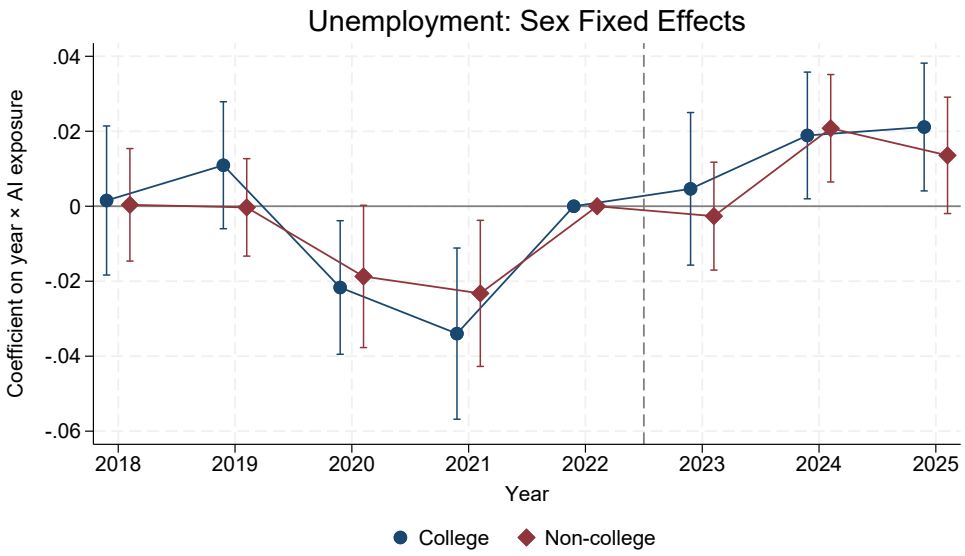

Figure B5: Unemployment: DiD coefficients, college (navy) vs. non-college (maroon), sex fixed effects. Results are nearly identical to the naive specification for both groups.

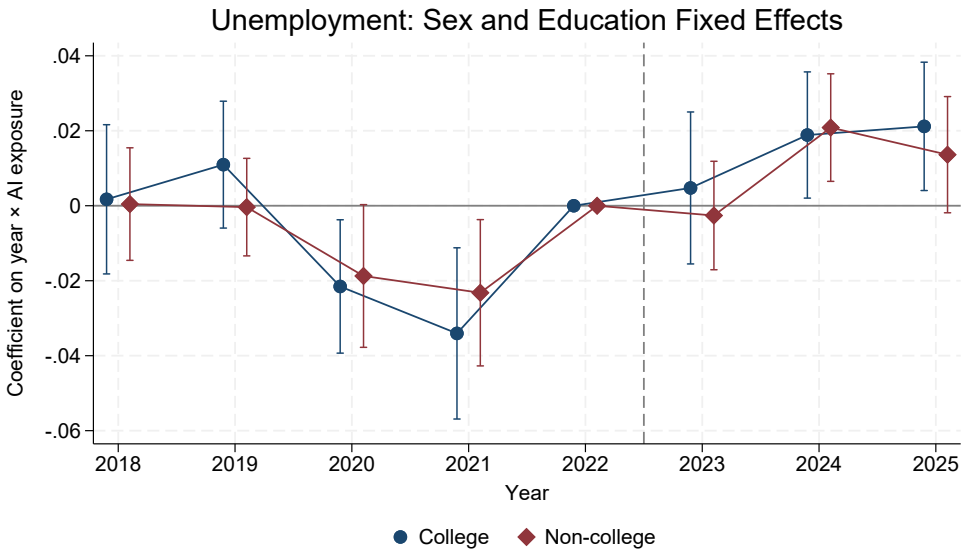

Figure B6: Unemployment: DiD coefficients, college (navy circles) vs. non-college (maroon diamonds), sex and education fixed effects. Year 2022 is the baseline. Error bars are 95% CI, clustered at the occupation level. Horizontal jitter ±0.1 years for readability.

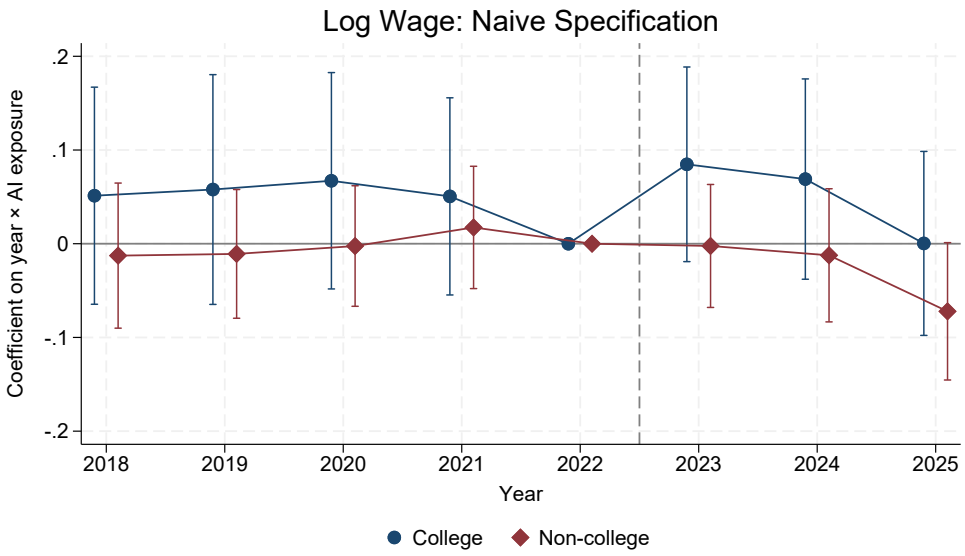

Figure B7: Log wage: DiD coefficients, college (navy circles) vs. non-college (maroon diamonds), naive specification. Year 2022 is the baseline. Error bars are 95% CI, clustered at the occupation level.

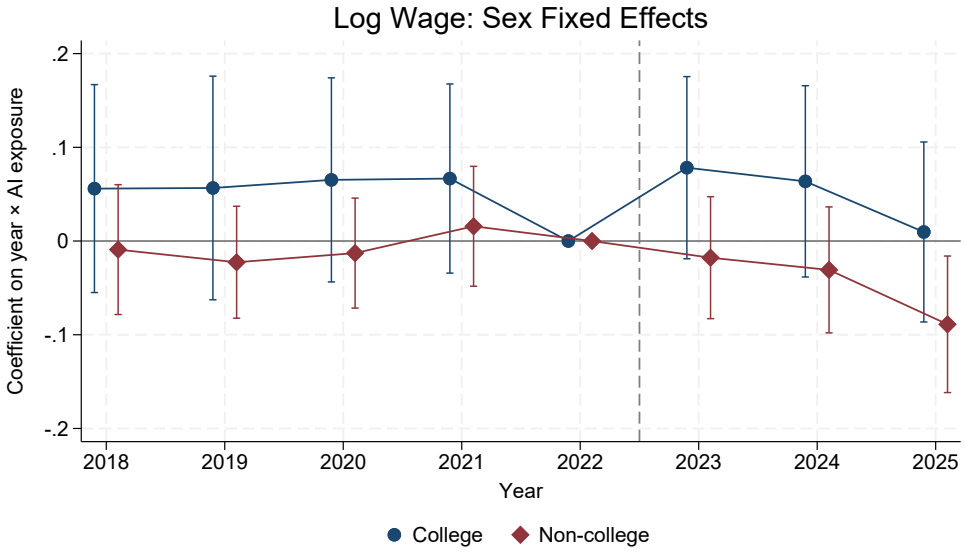

Figure B8: Log wage: DiD coefficients, college (navy) vs. non-college (maroon), sex fixed effects. The 2025 non-college divergence is already visible under this specification.

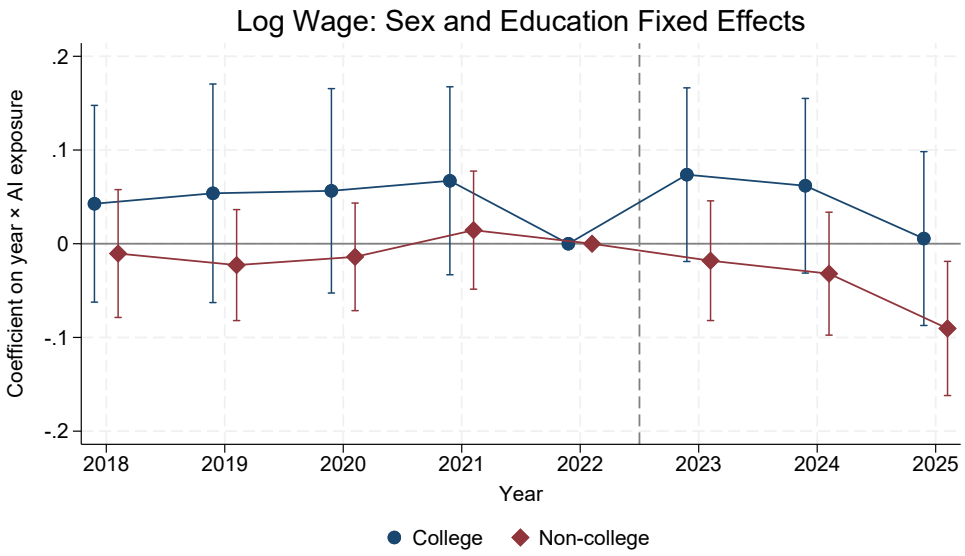

Figure B9: Log wage: DiD coefficients, college (navy circles) vs. non-college (maroon diamonds), sex and education fixed effects. Year 2022 is the baseline. The 2025 non-college compression (-0.090^{**}) is visible under this specification. Error bars are 95% CI, clustered at the occupation level.